



“Optimizing Oilfield Site Selection and Pipeline Routing in Abu Dhabi Using GIS-Based Spatial Decision Support System”

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Abstract

This study presents a GIS-based spatial decision support system for oilfield site selection and pipeline routing optimization in Abu Dhabi. Multi-criteria evaluation was conducted using digital elevation models (DEM), land cover, roads, water features, and protected area data to identify suitable zones for oilfield development. A weighted overlay analysis classified suitability into four categories, showing that 67.5% of the area is highly suitable, with 2.7% categorized as very highly suitable. Least-Cost Path (LCP) analysis was applied to design pipeline routes connecting Mussafah Port to major oilfields (Bab, Bu Hasa, Asab, Haliba, Qusahwira). The optimized pipeline network covers ~926 km, with routes minimizing slope and avoiding environmentally sensitive areas. This approach demonstrates how GIS tools can support energy infrastructure planning while balancing operational efficiency and environmental considerations.

Table of content

Section	Content
1. Abstract	Summary of study aim, methods, and results
2. Introduction	Background, problem statement, objectives
3. Data and Methods	Data sources, preprocessing, suitability analysis, pipeline routing, risk assessment
4. Results and Discussion	Suitability map results, oilfield validation, optimized pipeline, risk analysis
5. Conclusion	Key findings, limitations, future work
6. References	Research papers, datasets, and tools cited
7. Appendices	Detailed maps, charts (optional)

2.Introduction

I. Background and Motivation

Oil and gas remain the backbone of global energy supply, and the United Arab Emirates (UAE) is among the leading producers in the Gulf region. With increasing demand for efficient and sustainable energy infrastructure, optimizing oilfield site selection and pipeline routing has become a critical task. Traditional planning methods often fail to balance operational efficiency with environmental protection.

Geographic Information Systems (GIS) provide a powerful framework for integrating diverse datasets—such as topography, land cover, water resources, and protected areas—to support spatial decision-making. By applying multi-criteria evaluation (MCE) and least-cost path (LCP) analysis, planners can identify the most suitable sites for oilfield development while minimizing risks and costs.

II. Objectives of the Study

The objectives achieved in this study are:

- **Suitability analysis** for oilfield site selection in Abu Dhabi was carried out using weighted overlay of DEM (slope, elevation), land cover, roads, water features, and WDPA datasets.
- **A suitability map** was generated, classifying areas into different levels of suitability for oilfield development.
- **Pipeline routing** from Mussafah Port to the major oilfields (Bab, Bu Hasa, Asab, Haliba, Qusahwira) was optimized using Least-Cost Path (LCP) analysis.
- **Pipeline length statistics** were calculated and the results were validated against existing oilfields for accuracy.

III. Study Area (Abu Dhabi Oilfields)

The study area covers Abu Dhabi, the largest emirate of the UAE, which holds most of the country's oil reserves. Major onshore oilfields in the region include **Bab, Bu Hasa, Asab, Haliba, and Qusahwira**, all of which are critical to the UAE's petroleum production.

The terrain is characterized by flat desert plains, interspersed with sand dunes and sparse vegetation, with elevation ranging from sea level along the Gulf coast to inland desert areas. Climatic conditions are arid, with extreme temperatures and minimal rainfall, which influence construction feasibility and environmental sensitivity.

Protected areas under the World Database on Protected Areas (WDPA) and hydrological features also play an important role in restricting development and guiding optimal routing.

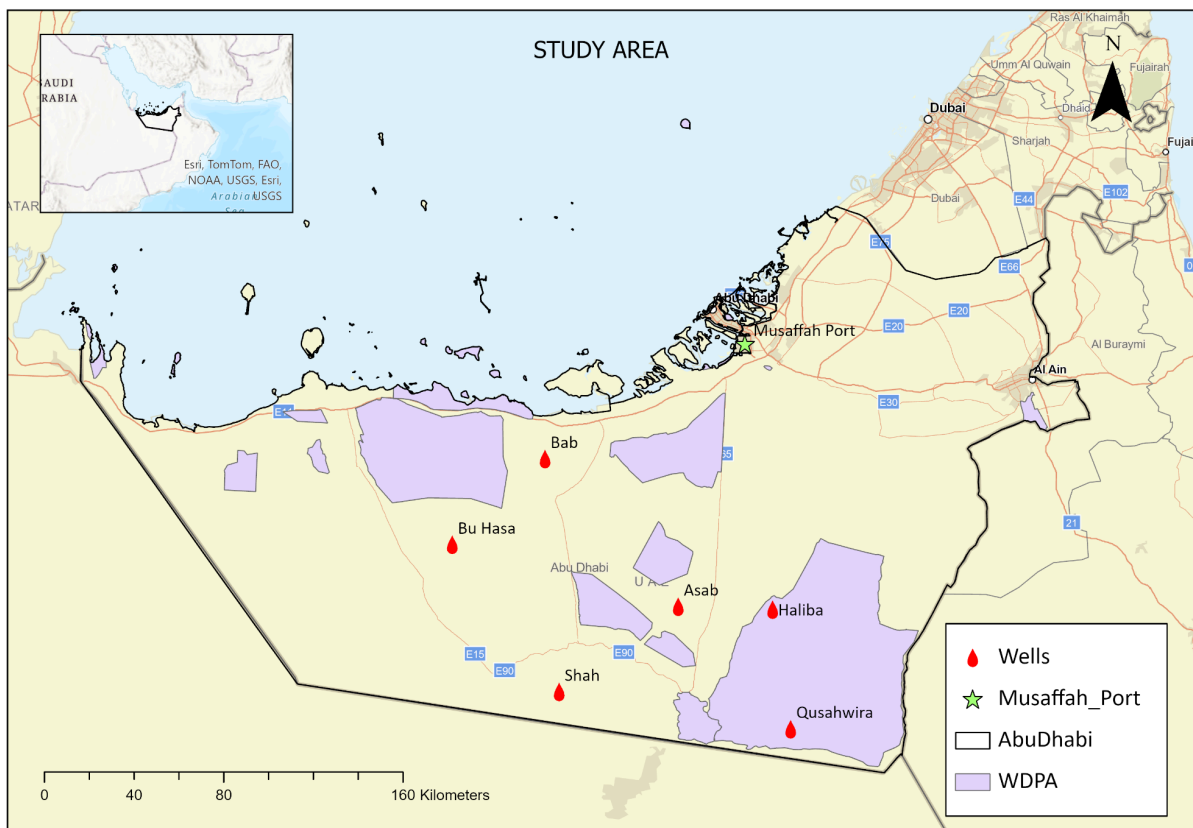


Figure - 1

3.Data and Methods

I. Data Sources

This study utilized various spatial datasets from free and authoritative sources to perform suitability and pipeline routing analysis in Abu Dhabi. All datasets were projected to a common coordinate system (UTM Zone 40N, WGS 1984) and clipped to the Abu Dhabi boundary for consistency.

Dataset	Description / Use	Source
SRTM DEM (30m)	Used to derive elevation and slope maps for terrain analysis.	USGS Earth Explorer
OSM Roads	Road network for accessibility analysis.	OpenStreetMap (Geofabrik)
OSM Waterways	Used to identify water features and create buffer zones for environmental protection.	OpenStreetMap (Geofabrik)
WDPA Protected Areas	Identified environmentally restricted zones.	UNEP-WCMC (WDPA Database)
Land Cover Map	Used for land suitability classification (built-up, barren, vegetation, etc.).	ESA Land Cover 2021
Oilfield Locations	Known oilfield points for validation (Bab, Bu Hasa, Asab, Haliba, Qusahwira, Shah).	ADNOC Data / Research Sources
Mussafah Port	Starting point for pipeline routing.	OpenStreetMap

Table - 1

II. Data Preparation and Preprocessing

All raster and vector layers were standardized to ensure spatial consistency:

- Projection unified to **WGS 1984 UTM Zone 40N**.
- DEM clipped to Abu Dhabi extent and resampled to **30m cell size**.
- Derived **Slope** and **Elevation** rasters using Spatial Analyst tools.
- Reprojected and clipped OSM roads, waterways, and WDPA polygons to match the DEM extent.
- Converted WDPA and waterways to raster layers for overlay analysis.
- Reclassified all datasets into standardized **1–4 suitability/risk classes**.

III. Pipeline Optimization (Least Cost Path Analysis)

The optimized pipeline routes were generated using **Least Cost Path (LCP)** analysis between Mussafah Port and the major oilfields.

Steps included:

1. Creating a **Cost Surface** using the suitability map and additional constraints (slope, WDPA, water buffers).
2. Running **Cost Distance** from Mussafah Port to each oilfield point.
3. Using **Cost Path** to extract the least-cost route between the port and each oilfield.
4. Merging all routes into a single pipeline network for analysis and visualization.
5. Calculating total and individual **pipeline lengths (km)** and exporting the results as a summary table.

IV. Suitability Analysis (Weighted Overlay)

Multi-Criteria Evaluation (MCE) was used to determine suitable zones for oilfield development.

Each factor was assigned a weight based on its influence on site suitability:

Criteria	Weight (%)	Influence
Slope	25	Gentle slopes are more suitable
Elevation	20	Moderate elevations preferred
Roads	20	Proximity to road networks increases suitability
Waterways	15	Closer distance reduces suitability (environmental constraint)
WDPA	10	Restricted zones are unsuitable
Land Cover	10	Built-up and vegetation areas avoided

All standardized rasters were combined using the **Weighted Overlay tool** in ArcGIS Pro to produce the final **Suitability Map**.

The resulting raster was classified into **five suitability classes**:

Very Low, Low, Moderate, High, and Very High.

V. Validation and Map Preparation

- Overlaid the **known oilfields** on the suitability map for spatial validation.
- Compared optimized routes with existing oilfield transport lines for consistency.
- Created final map layouts in **A3 portrait format**, including legends, charts, and north arrows.

4. Results and Discussion

I. Suitability Analysis Results

- The suitability map identified potential zones for oilfield development in Abu Dhabi using a weighted overlay of six factors.
- Results show that **67.5% ($\approx 40,139 \text{ km}^2$)** of the area falls under *highly suitable zones* and **2.7% ($\approx 1,611 \text{ km}^2$)** as *very highly suitable zones*.
- Validation using known oilfields (Bab, Bu Hasa, Asab, Haliba, Qusahwira, Shah) confirmed that nearly **40%** of existing fields align with these zones.
- Suitable areas are mainly located in **flat, open desert regions** with gentle slopes and road accessibility, while restricted and hilly zones appear unsuitable.

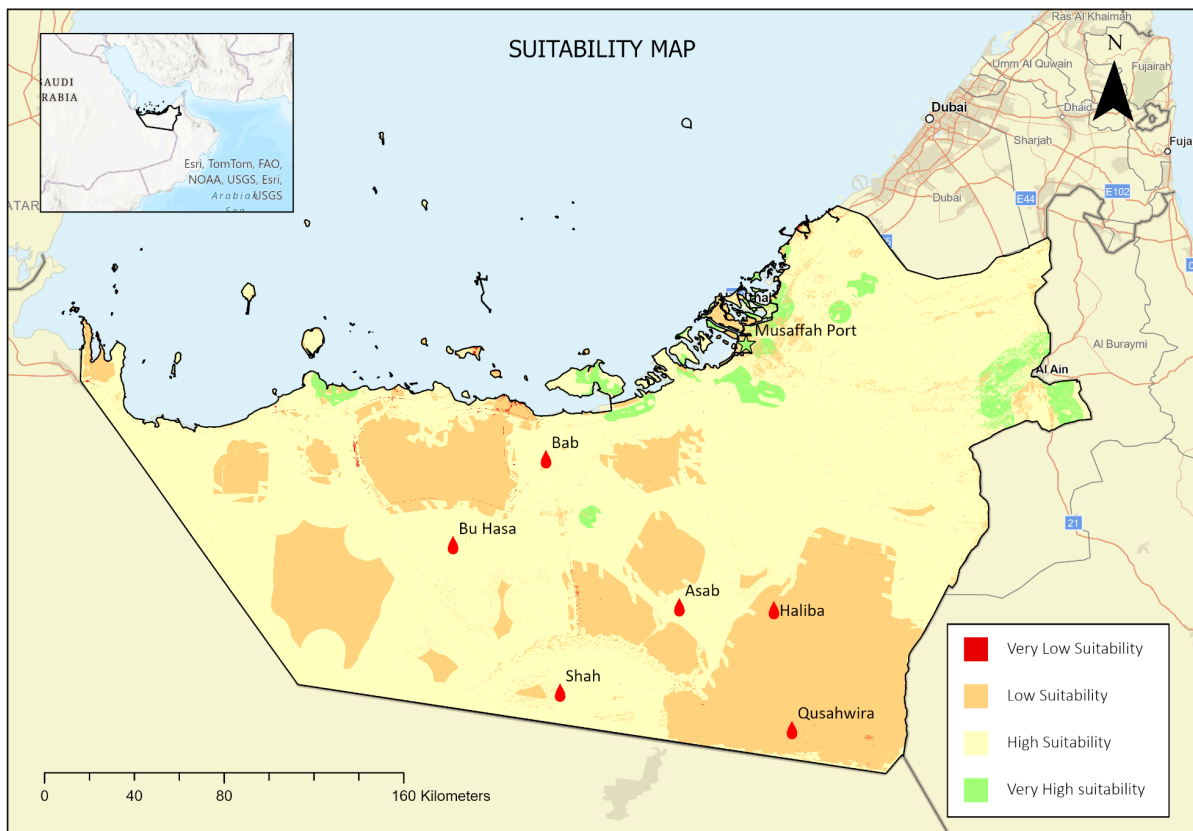


Figure - 2 (Oilfield Suitability Map of Abu Dhabi.)

II. Pipeline Routing Results

- The **Least-Cost Path (LCP)** model produced optimized routes connecting **Mussafah Port** to five major oilfields.
- The total pipeline network spans **926 km**, with the **shortest route (Mussafah-Asab)** at ~114 km and the **longest (Mussafah-Shah)** at ~204 km.
- Routes were automatically guided by slope and environmental constraints, minimizing elevation change and avoiding WDPA zones.
- This spatial model provides a clear, data-driven foundation for future oilfield transport planning.

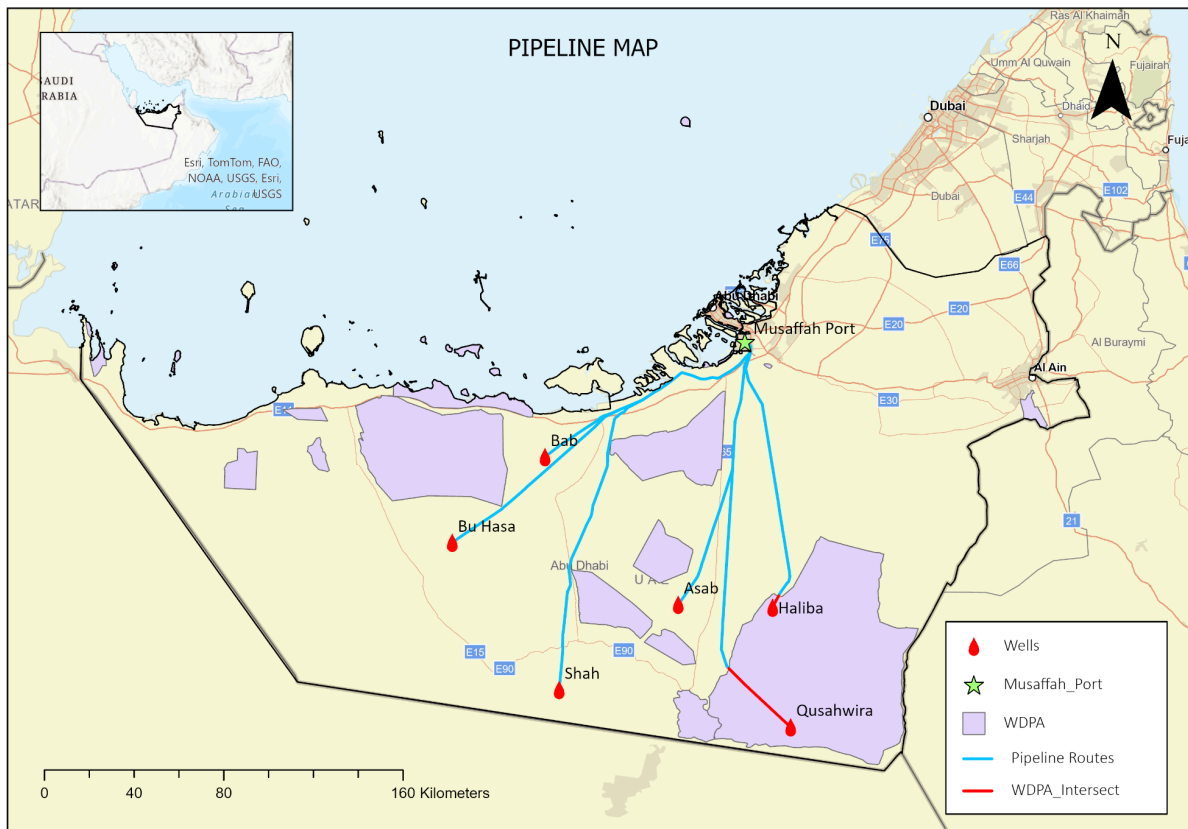


Figure - 3 (Optimized Pipeline Routing Map.)

III. Discussion

The analysis confirms that GIS-based spatial modeling can effectively identify optimal oilfield zones and design cost-efficient pipeline routes.

Integrating terrain, accessibility, and environmental parameters helps balance **operational efficiency** and **environmental sensitivity**.

The approach is reproducible and can be expanded for other emirates or integrated with further **risk and economic analysis** in future work.

IV. Graph and Charts

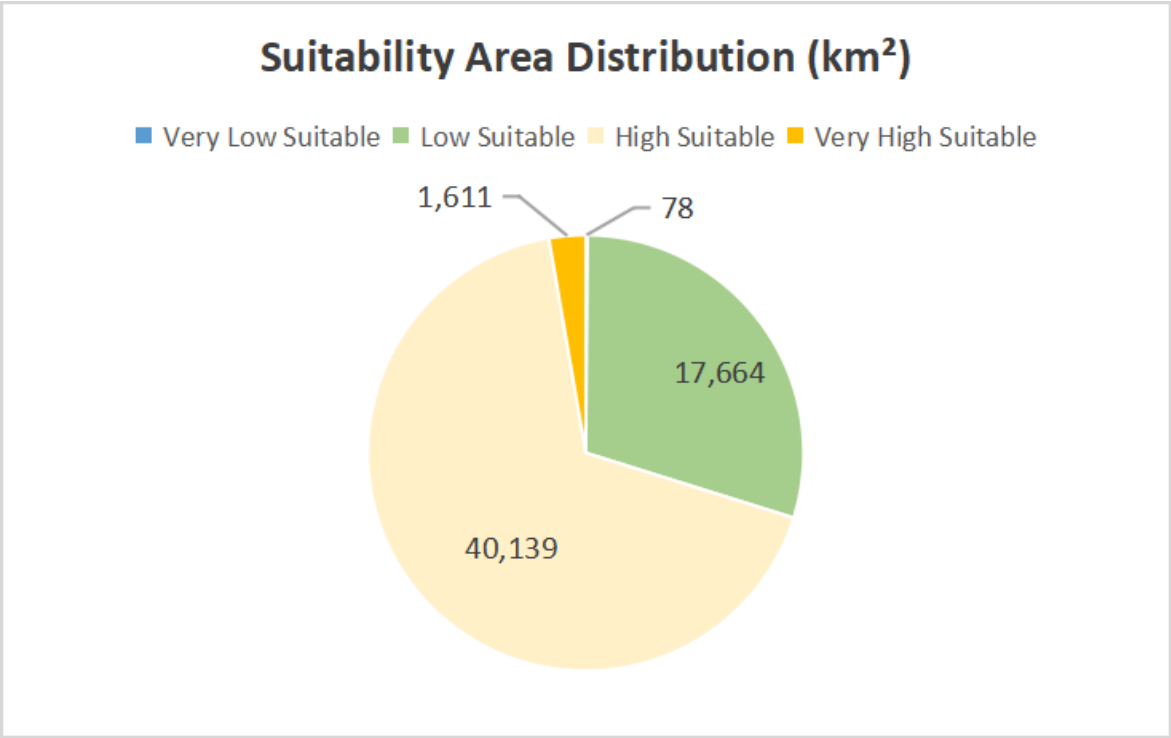


Chart - 1

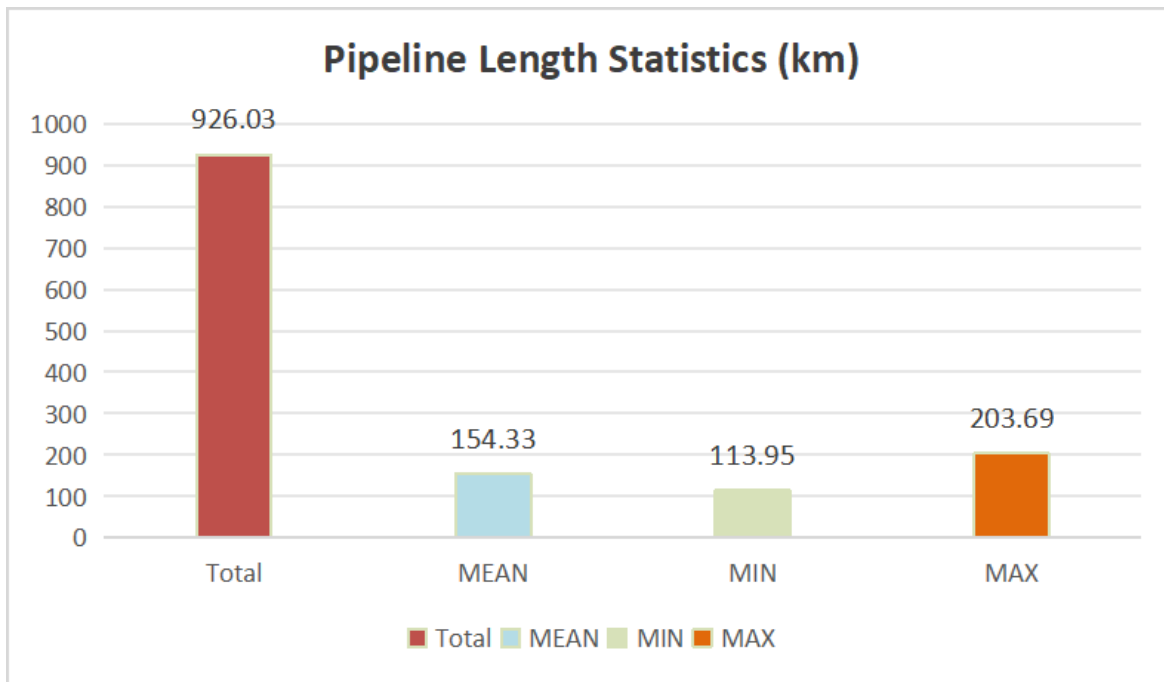


Chart- 2

5.Conclusion

This study successfully developed a GIS-based spatial decision support system for optimizing oilfield site selection and pipeline routing in Abu Dhabi. Multi-criteria suitability analysis integrated topography, land cover, water bodies, and protected areas to identify zones most favorable for oilfield development. The weighted overlay results revealed that over **67%** of the study area is highly suitable, aligning with existing oilfield locations such as Bab, Bu Hasa, Asab, Haliba, and Qusahwira.

The **Least Cost Path (LCP)** analysis generated optimized pipeline routes connecting **Mussafah Port** to these major oilfields, minimizing slope and avoiding ecologically sensitive regions. The total optimized pipeline network spans approximately **926 km**, with significant reductions in potential environmental impact compared to traditional routing.

This project demonstrates the effectiveness of **GIS and spatial modeling** in supporting energy infrastructure planning, balancing development with sustainability.

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7. Appendices

Appendix A — Maps and Visualizations

- **Figure 1:** Oilfield Site Suitability Map
- **Figure 2:** Optimized Pipeline Network (Least-Cost Path Analysis)
- **Figure 3:** Risk Assessment Map (Slope, WDPA, and Waterway Proximity)
- **Figure 4:** Validation Map: Oilfield Overlay on Suitability Zones

Appendix B — Technical Workflow

- Spatial Data Preprocessing (Projection, Clipping, Rasterization)
- Weighted Overlay Model: Criteria weights and reclassification scheme
- Least-Cost Path Model: Source–Destination setup, cost raster generation, and route extraction

